Lecture notes for *Principles of Artificial Intelligence* (COMS 4720/5720)
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Search with Zero or Partial Percepts

The agent may not know the current state in a partially observable environment.

Outline

I. Searching with no observation

II. Searching in partially observable environments

III. Online search agents and unknown environments

* Figures are from the textbook site (or drawn by the instructor) unless the source is specifically cited.

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Agent's precepts provide no information at all.

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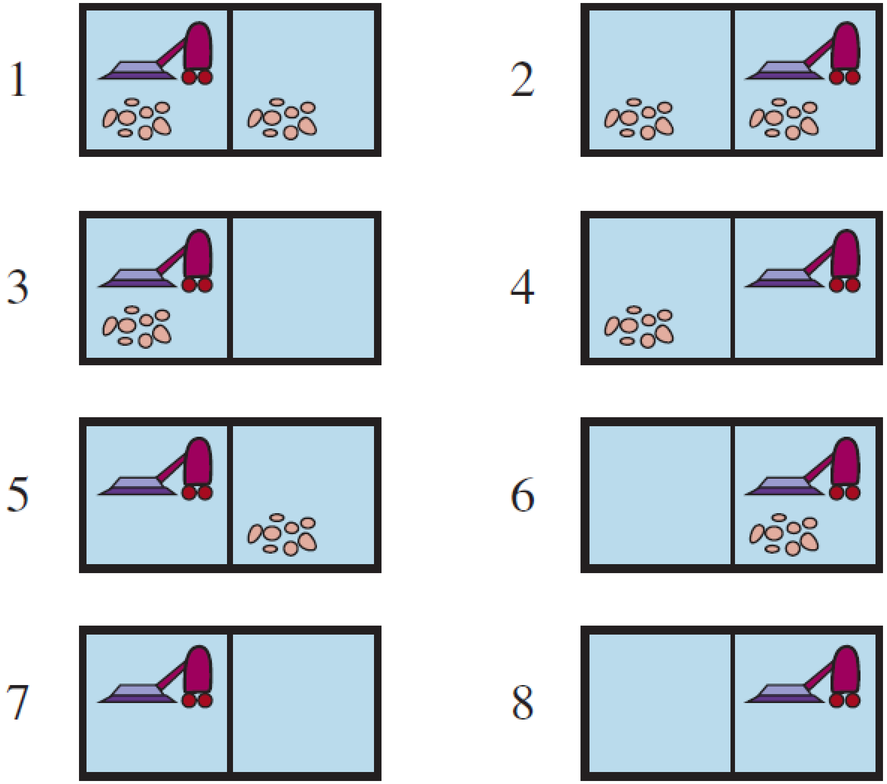

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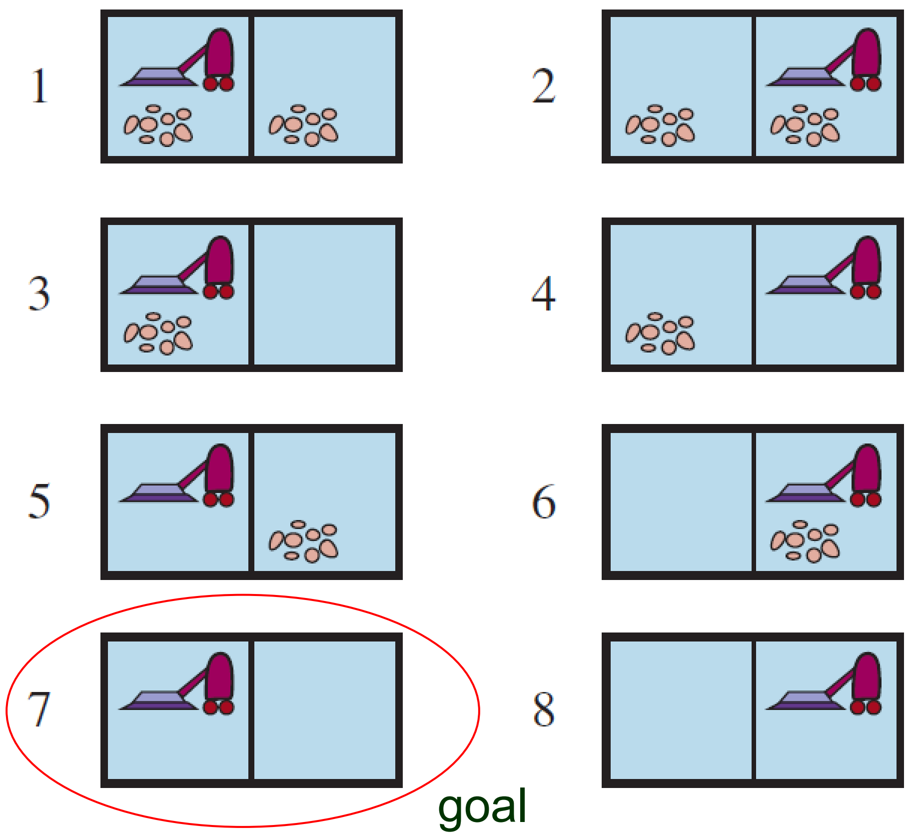

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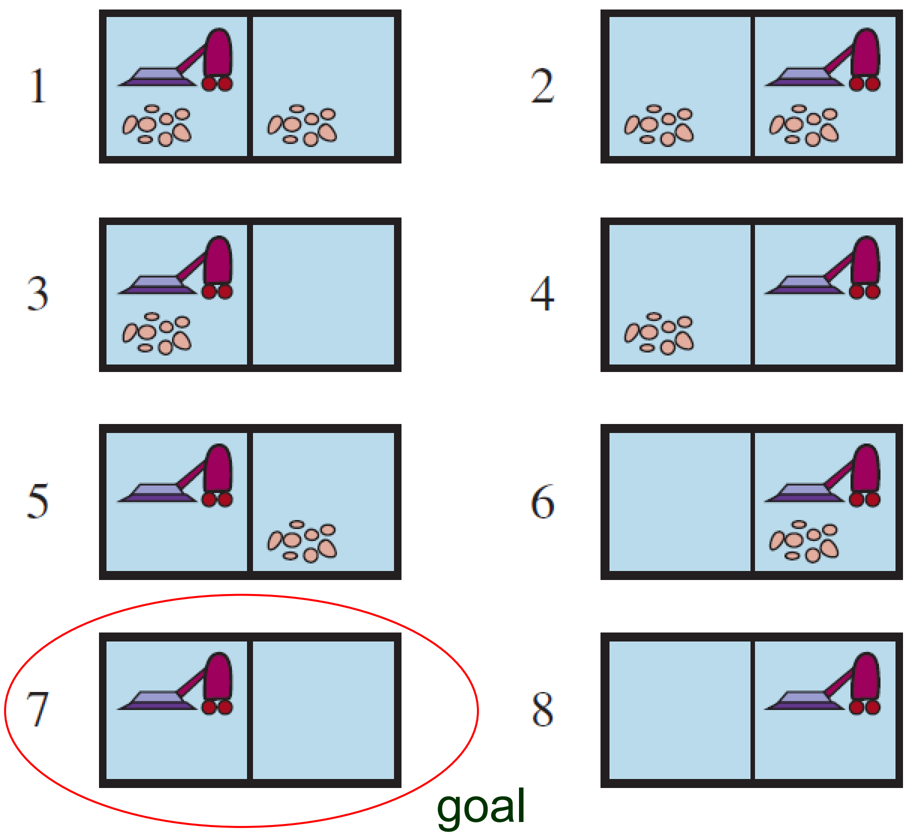

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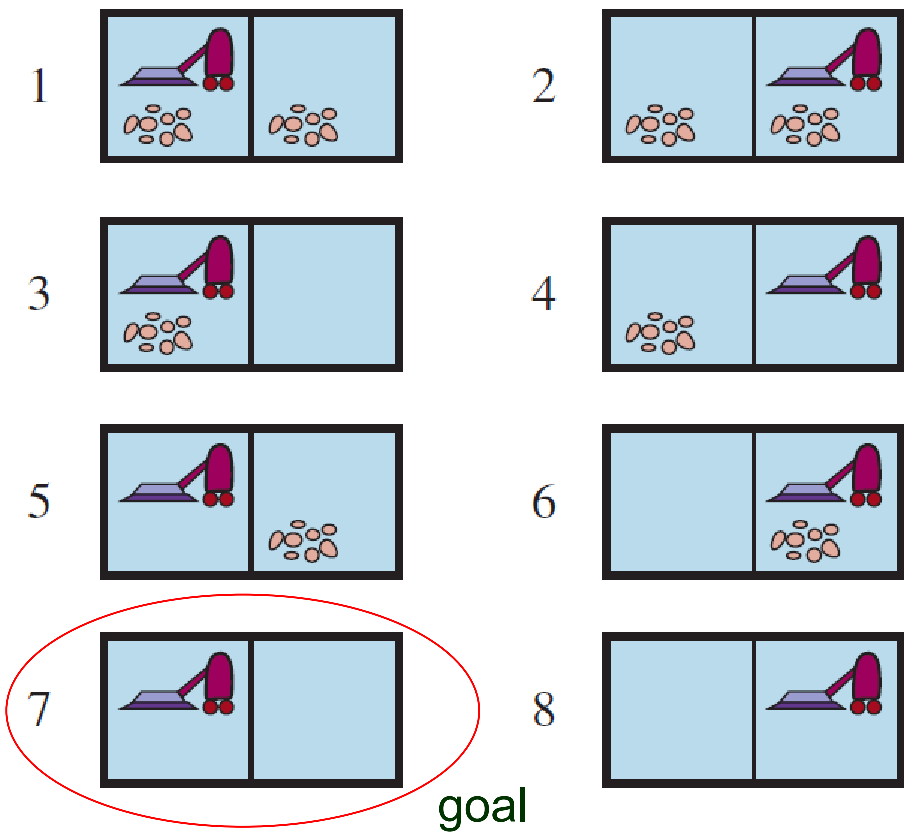

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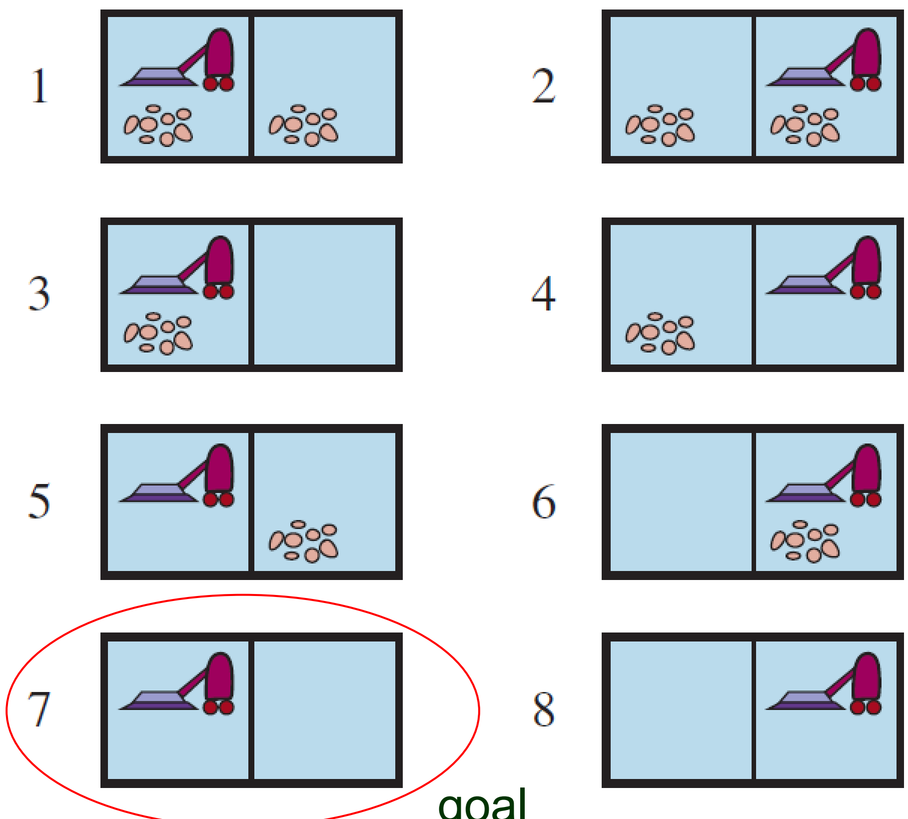

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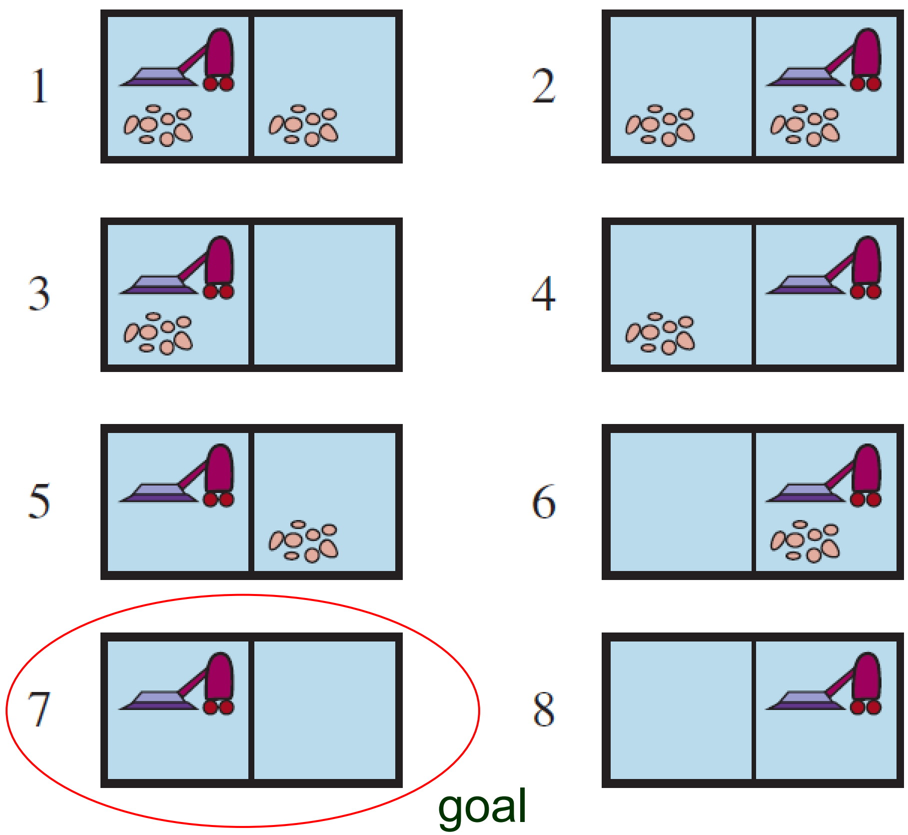

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Sensorless Manipulation – Tray Tilting

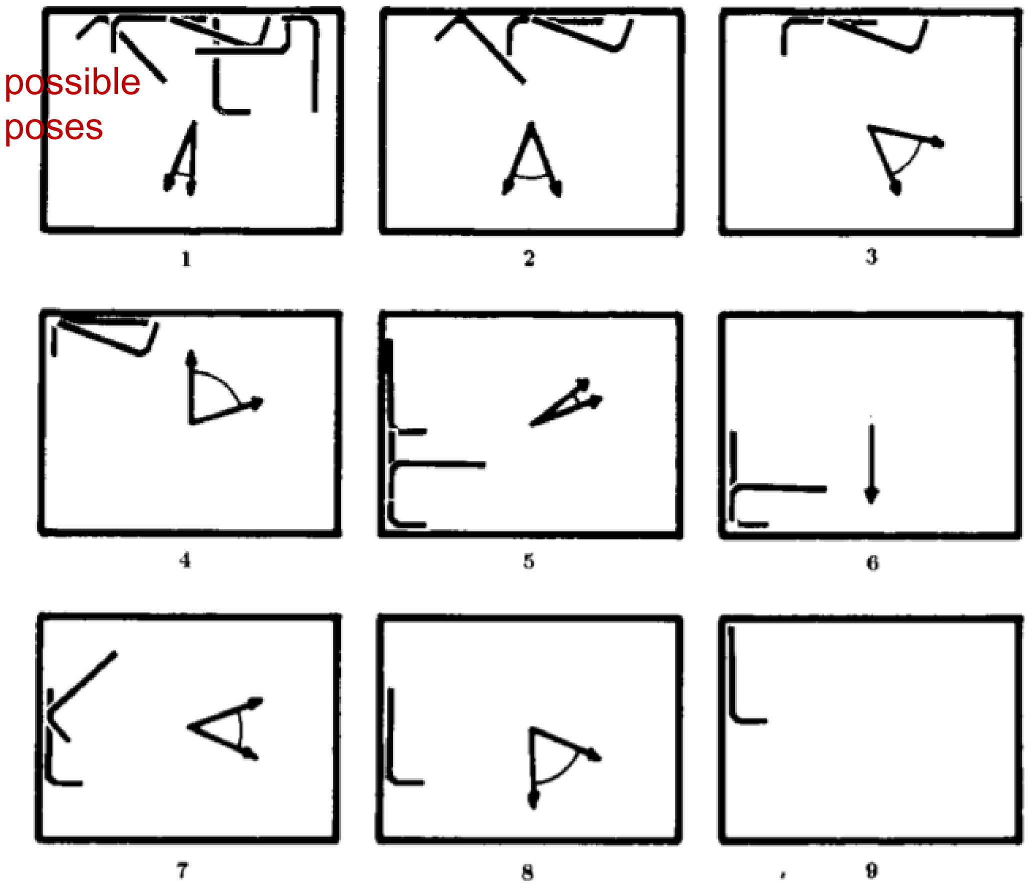

Fig. 2. Beginning at the upper left and moving from left to right, we can trace an automatically generated program that orients the wrench. Each frame shows the set of possible wrench contacts, and the operation to be applied. Each operation is represented by an interval of azimuths. The azimuth arrows indicate the tray's direction of steepest ascent; gravity acts in the opposite direction.

Erdmann & Mason (1988):

Orients the Allen wrench with eight tilts.

- Slow planar motion
- Coulomb friction

IEEE Journal of Robotics and Automation, vol. 4, no. 4, pp. 369-379, 1988.

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Action on a physical state

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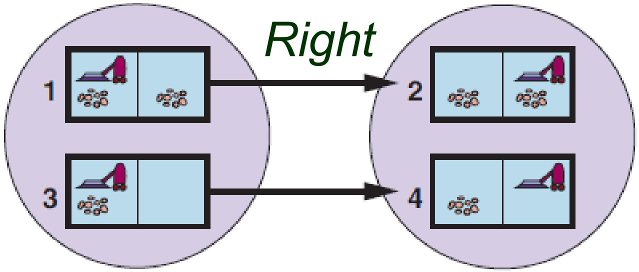

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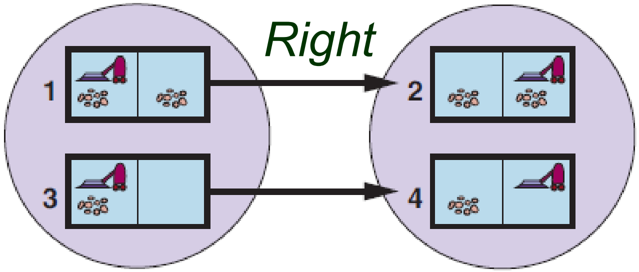

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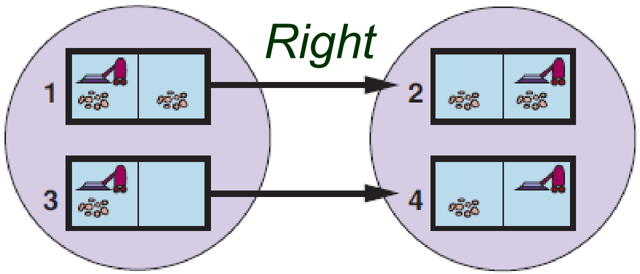

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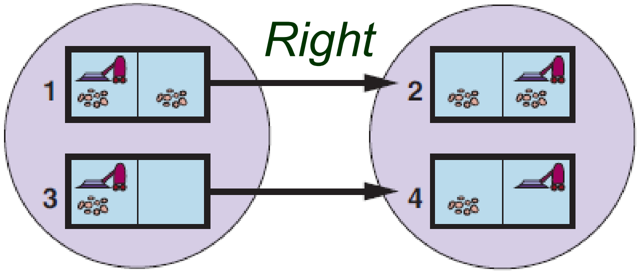

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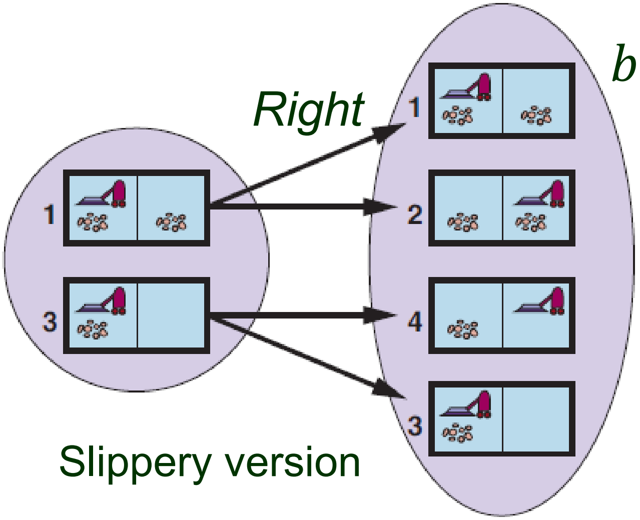

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Slippery version

Goal & Cost of Action

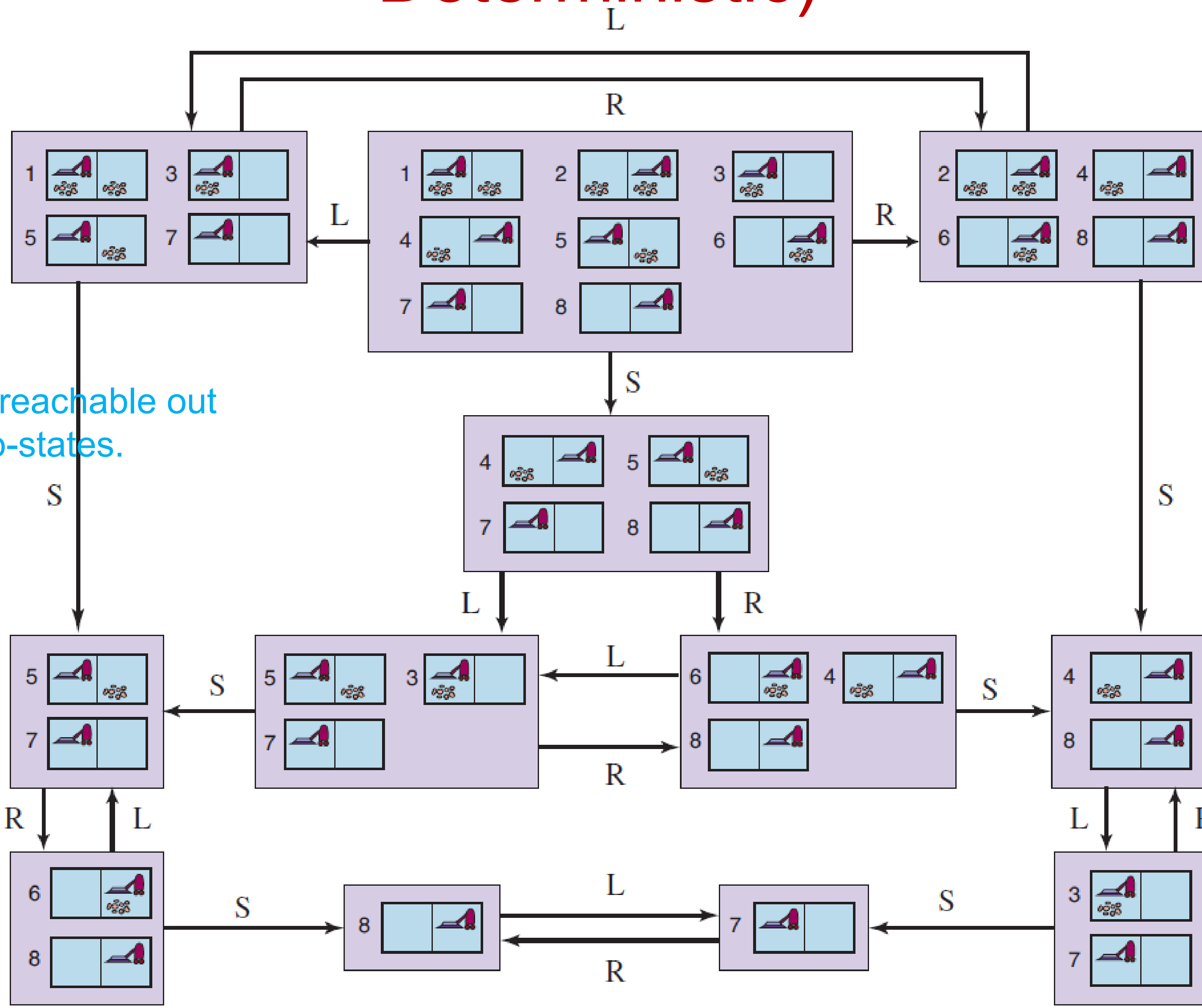
- Goal test: The goal is achieved

  - ♣ *possibly* if one of the states $s \in b$ passes the test;

  - ♣ *necessarily* if every state $s \in b$ passes the test.

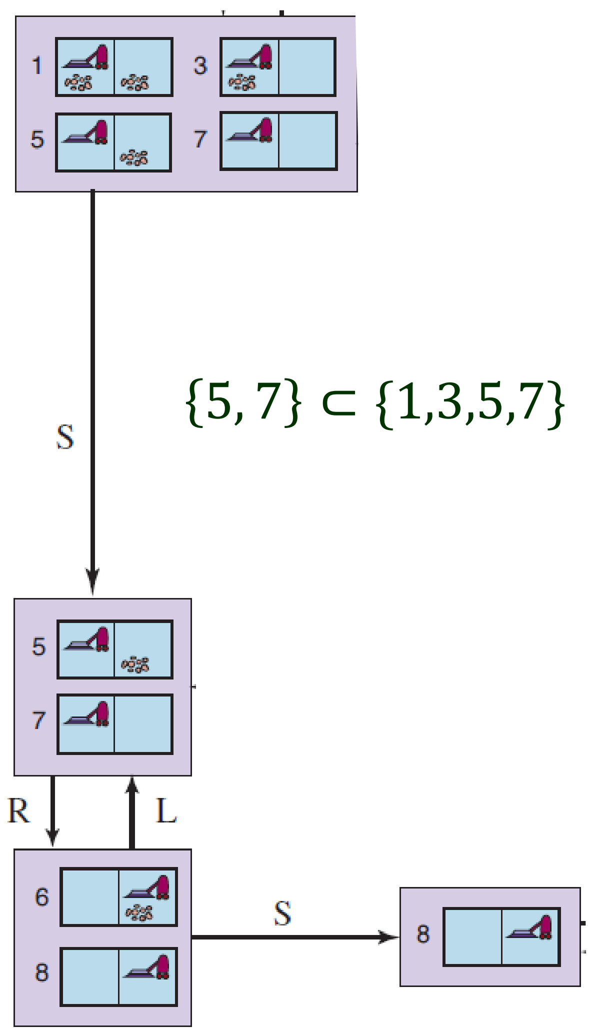
- Action cost: Could be one of several values if the same action has different costs in different states.

Reachable Belief States (Sensorless & Deterministic)



Only 12 are reachable out of 256= 2^8 b-states.

Subset and Superset States



- Prune a superset b-state, say, $\{1, 3, 5, 7\}$, to concentrate on solving the easier subset b-state, say, $\{5, 7\}$.
- Prune a subset b-state, say, $\{2, 5, 7\}$, if a superset b-state, say, $\{1, 3, 5, 7\}$, of the resulting subset, say, $\{5, 7\}$, has been found to be solvable already.

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    - If no, find a different solution for state 1, and so on.

II. Partially Observable Environments

Many problems (e.g., the 8-puzzle) are unsolvable without sensing.

A little sensing (e.g., only one visible square in the 8-puzzle) can go a long way.

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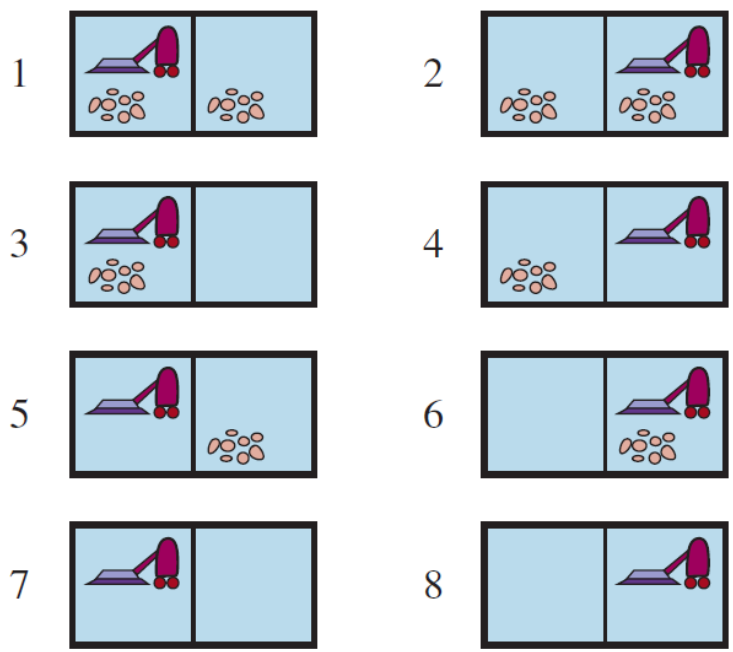

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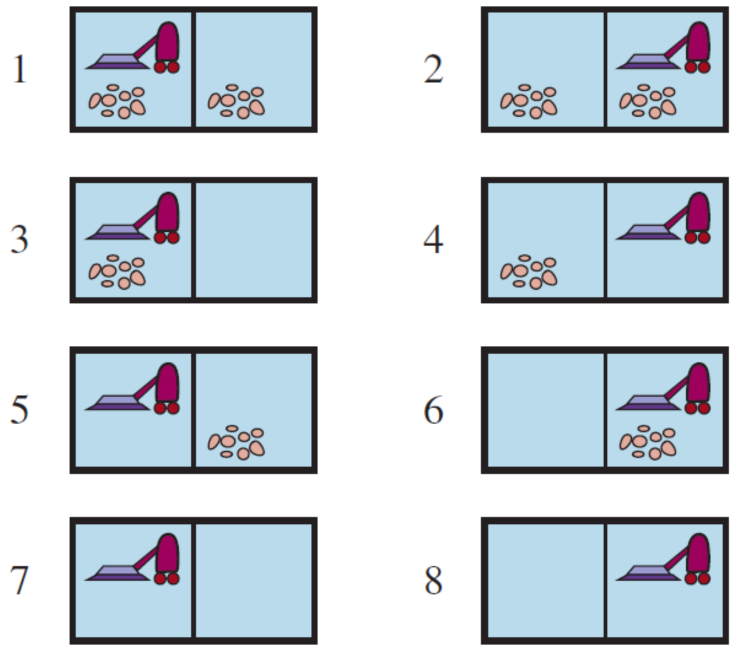

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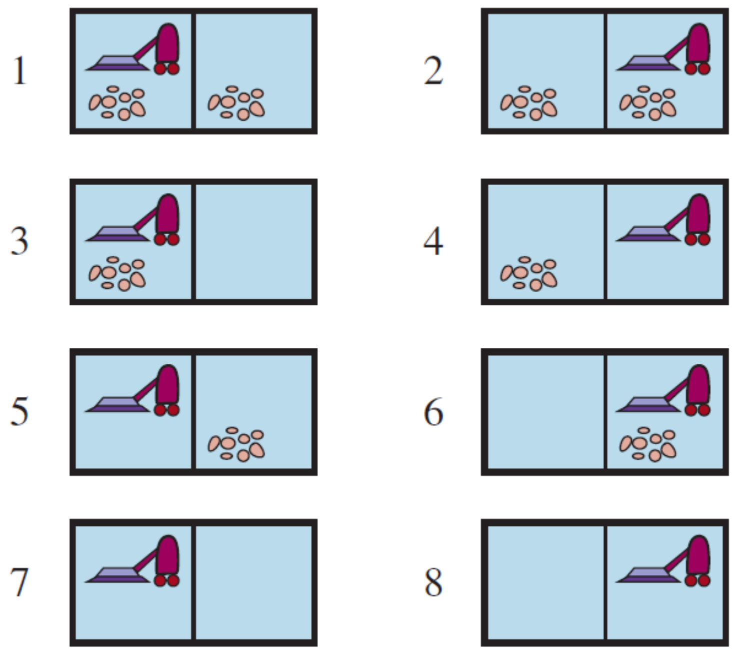

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$\{L, Dirty\} \longrightarrow$ b-state {1, 3}

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- Prediction: computes the b-state from an action a.

estimate

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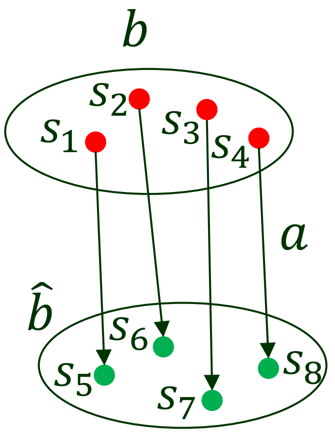

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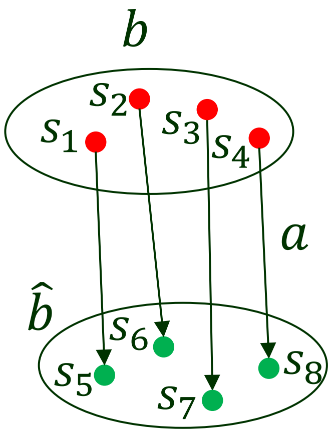

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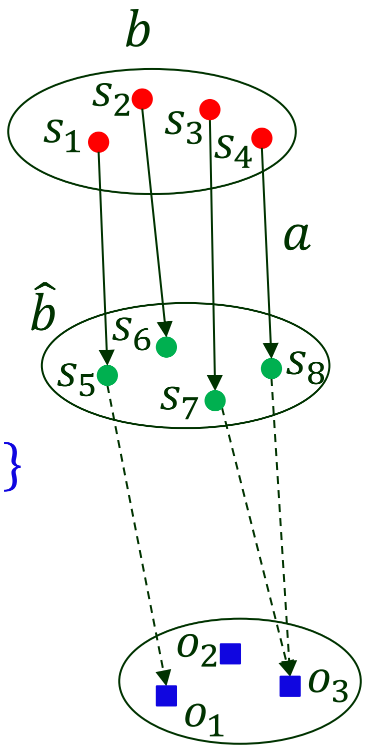

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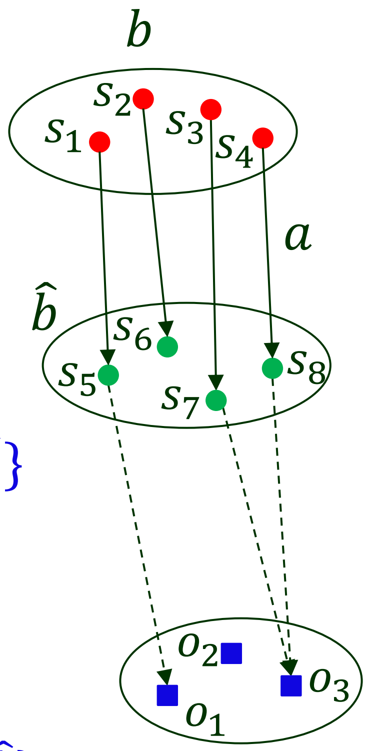

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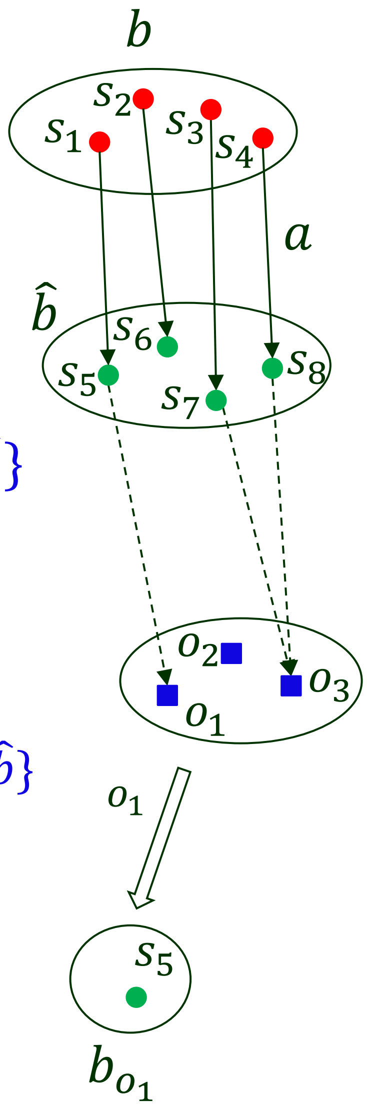

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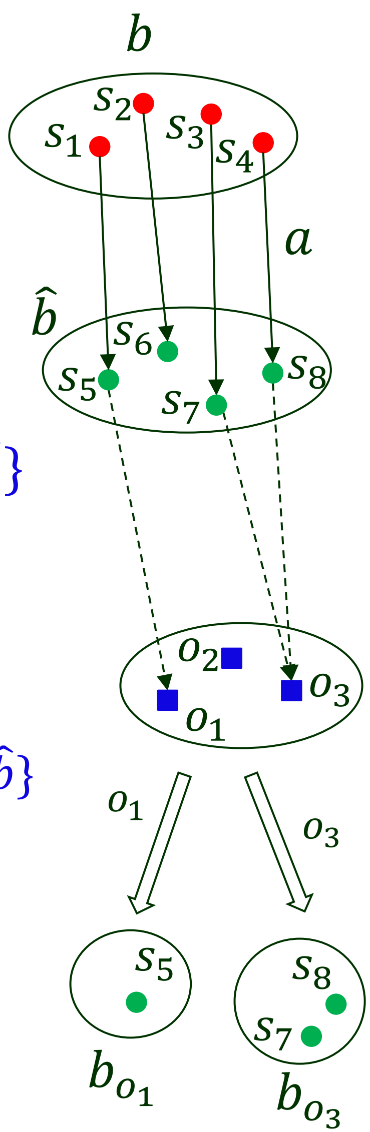

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- Result : resulting belief states
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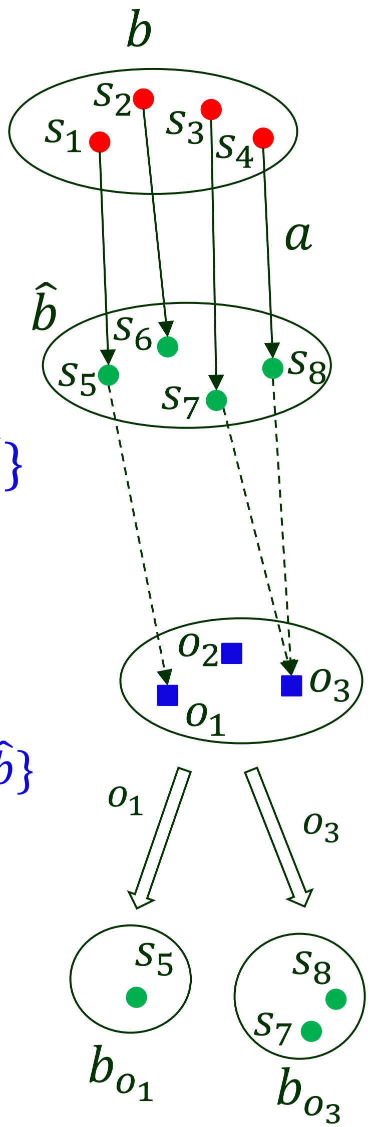

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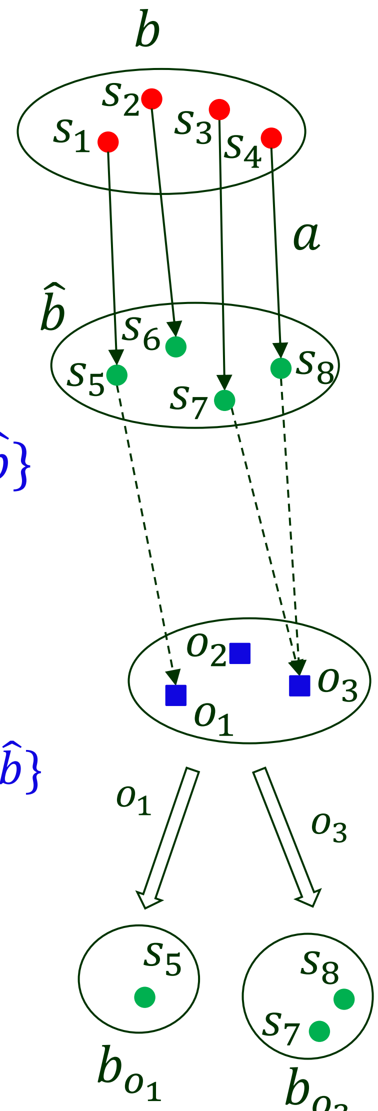
$$b_o = \text{UPDATE}(\hat{b}, o) = \{s \mid o = \text{POSSIBLE-PERCEPTS}(s) \text{ and } s \in \hat{b}\}$$

e.g., $b_{o_1} = \text{UPDATE}(\hat{b}, o_1) = \{s_5\}$ $b_{o_3} = \text{UPDATE}(\hat{b}, o_3) = \{s_7, s_8\}$

- Result : resulting belief states

$$\text{RESULT}(b, a) = \{b_o \mid o \in \text{POSSIBLE-PERCEPTS}(\hat{b})\}$$

e.g., $\text{RESULTS}(b, a) = \{b_{o_1}, b_{o_3}\}$



Possible B-states from an Action

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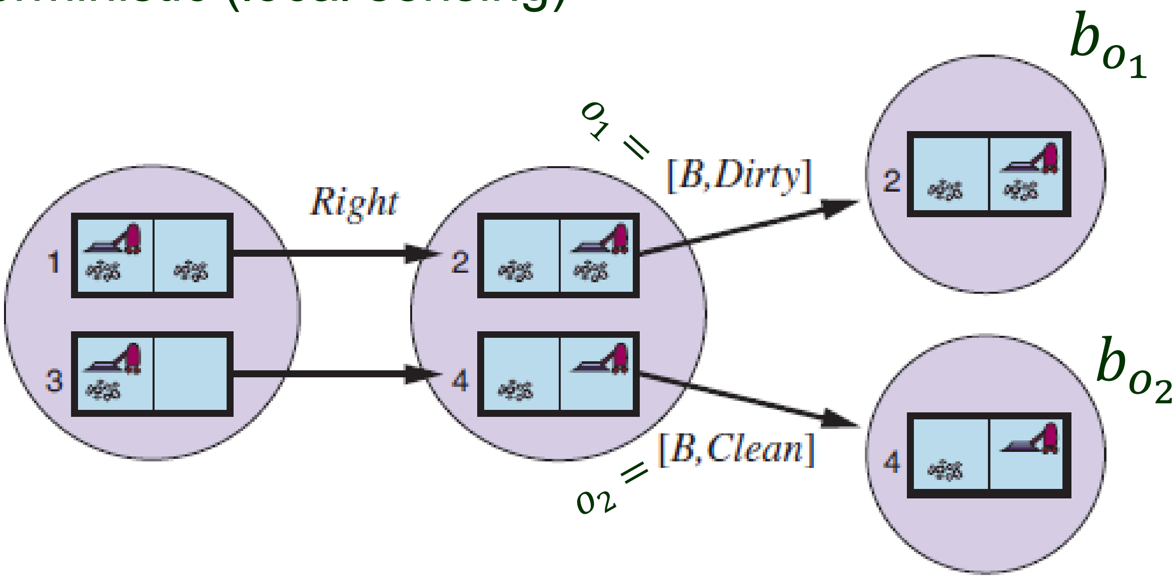
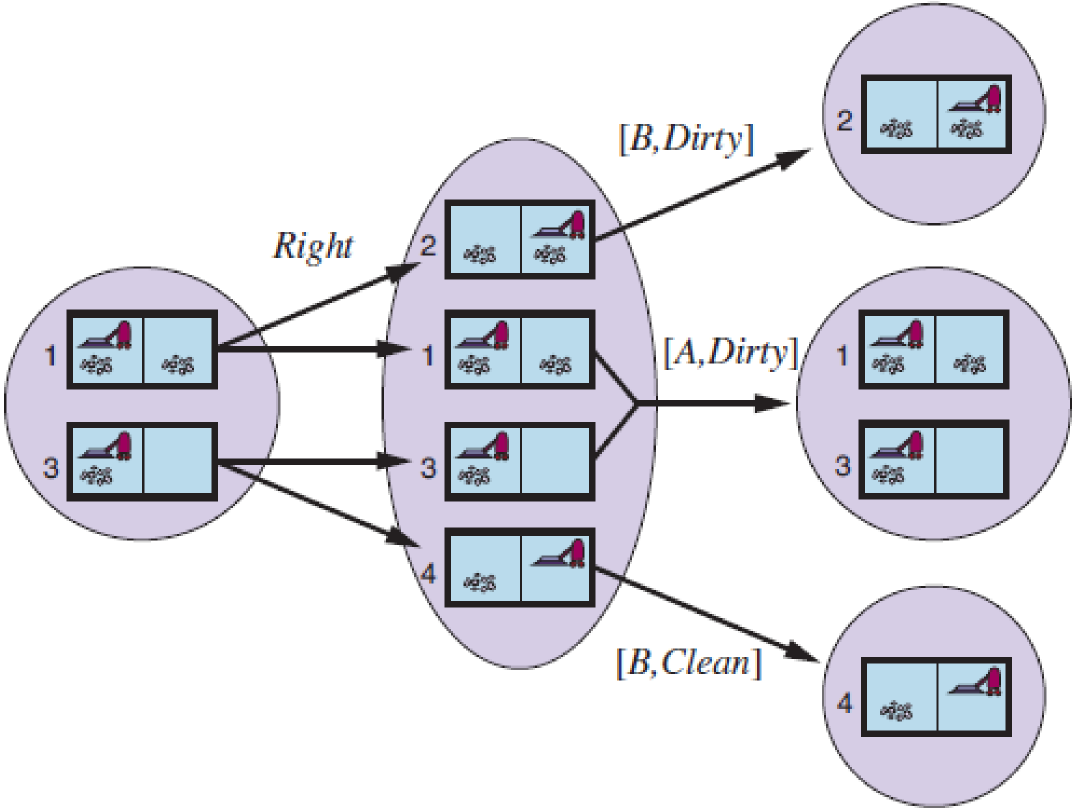


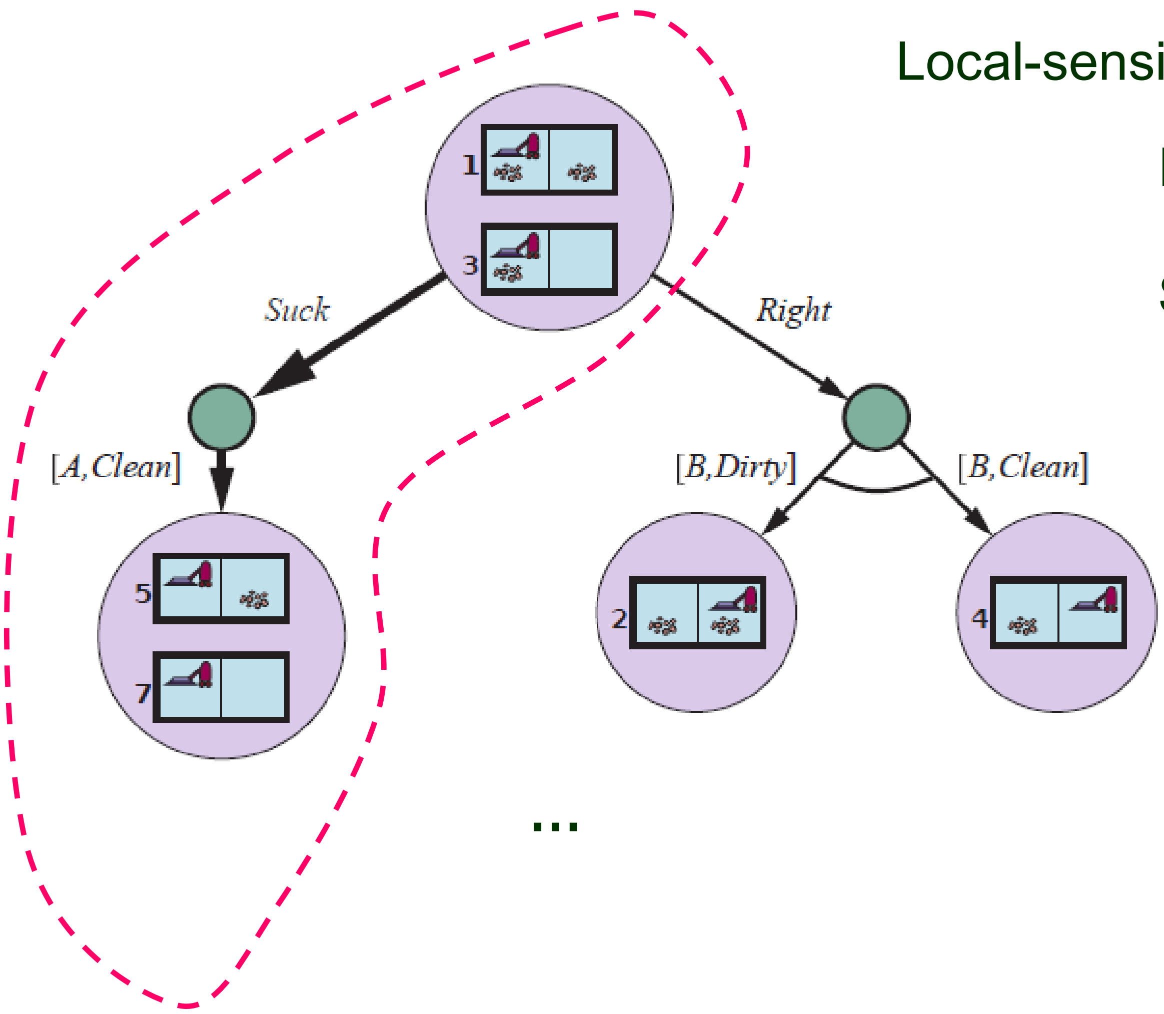
(cont'd)

- Nondeterministic (local sensing + slippery world)



Solution with Partial Observation

Apply the AND-OR search algorithm.



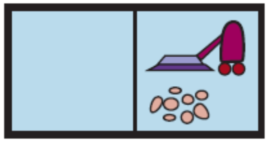
Local-sensing vacuum world

Initial percept: [*A*, *Dirty*]

Solution:

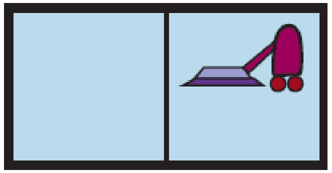
Suck
Right
if b-state={6}
then *Suck*
else []

6



Goal:

8



Prediction-Update Cycles

- Test the condition and execute the appropriate branch.
- Maintain its belief state as it performs actions and receives percepts.

$$b' = \text{U}\small{\text{PDATE}}\normalsize(\text{P}\small{\text{REDICT}}\normalsize(b, a), o)$$

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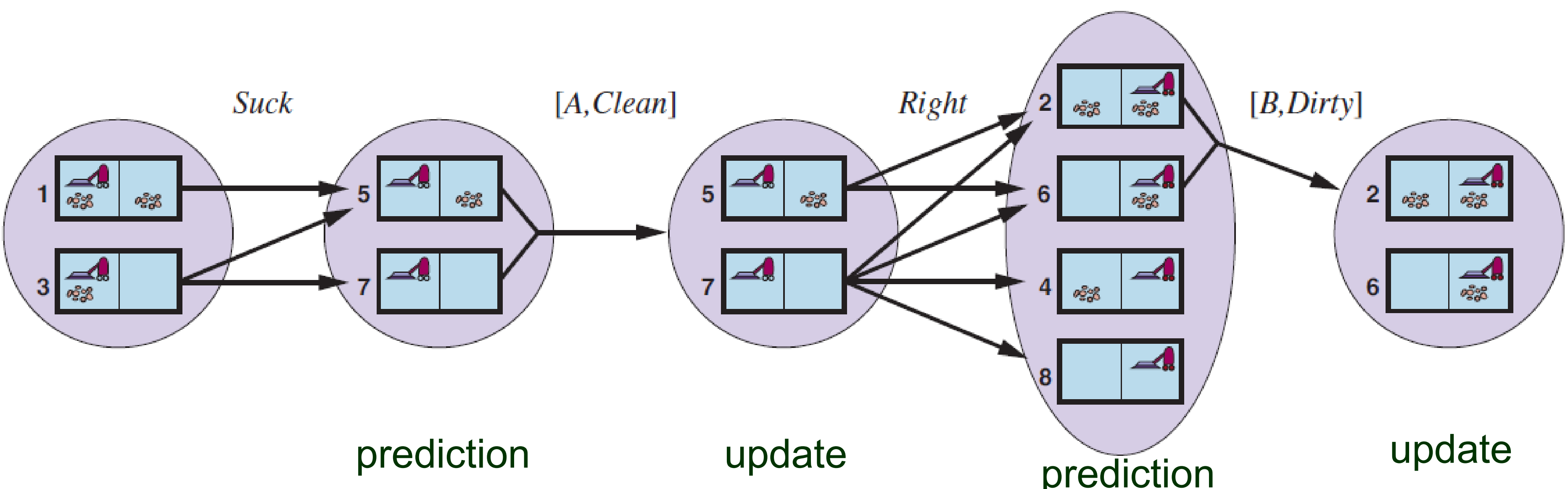

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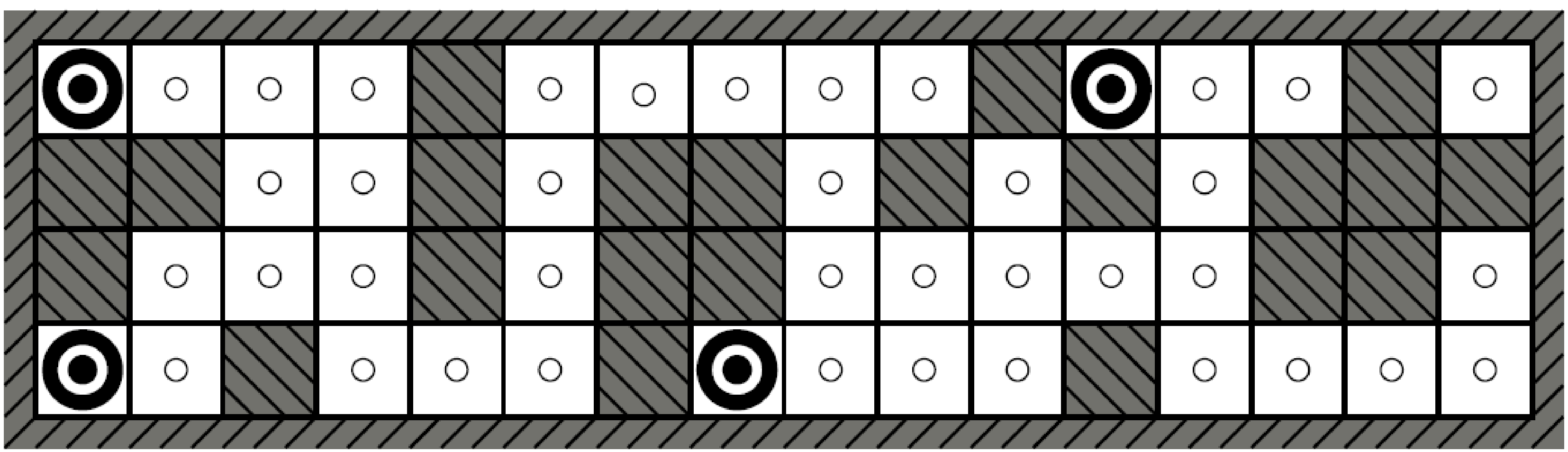

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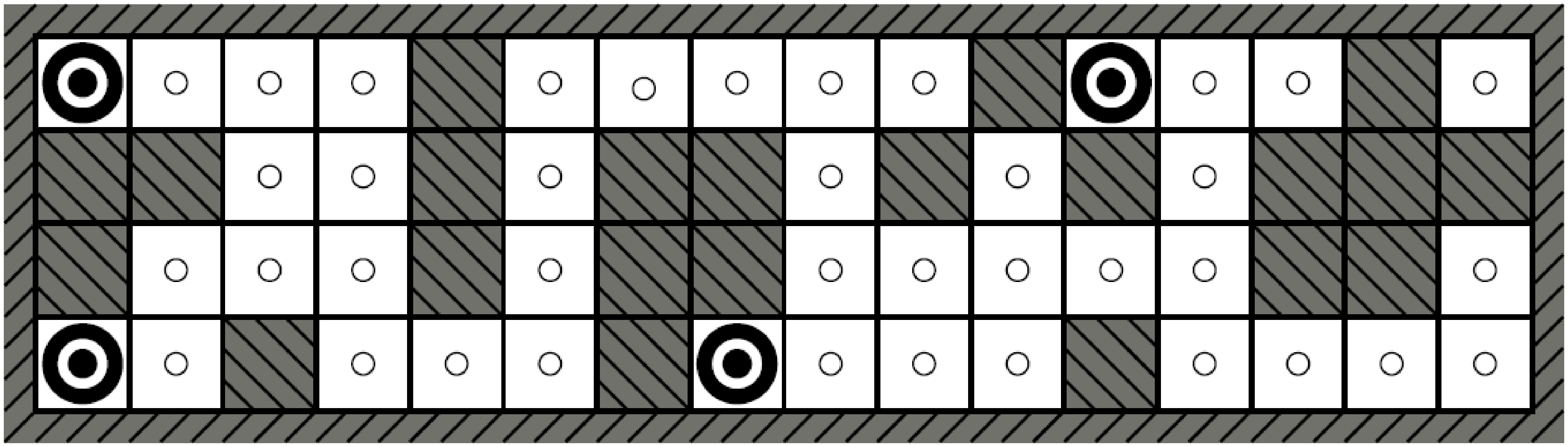

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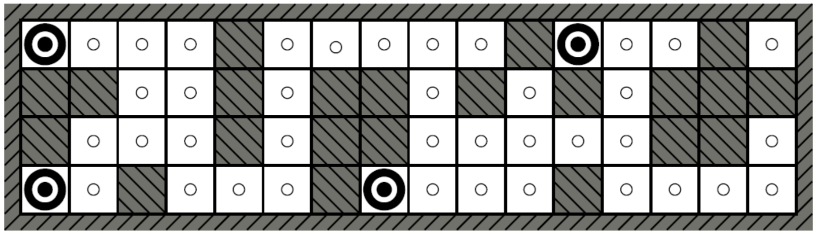

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Next, the robot executes a *Right* action.

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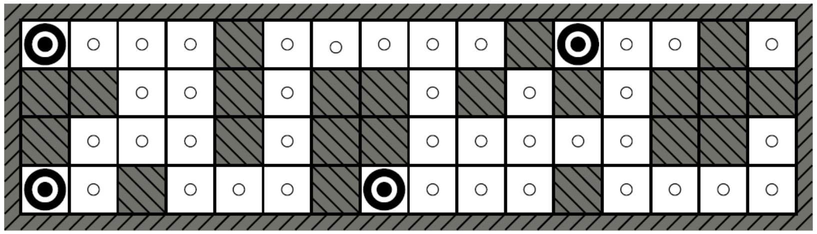

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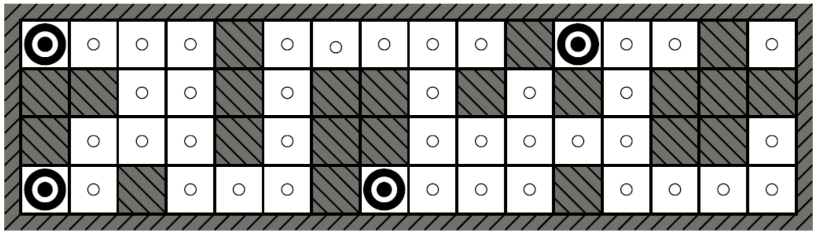

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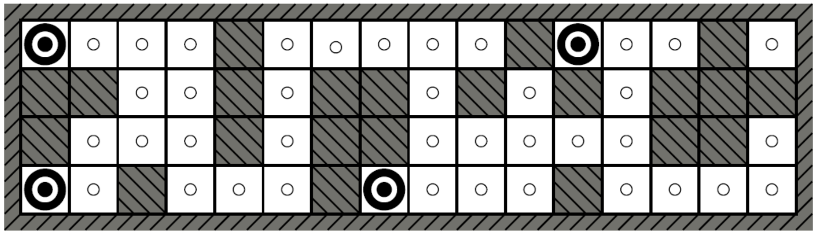
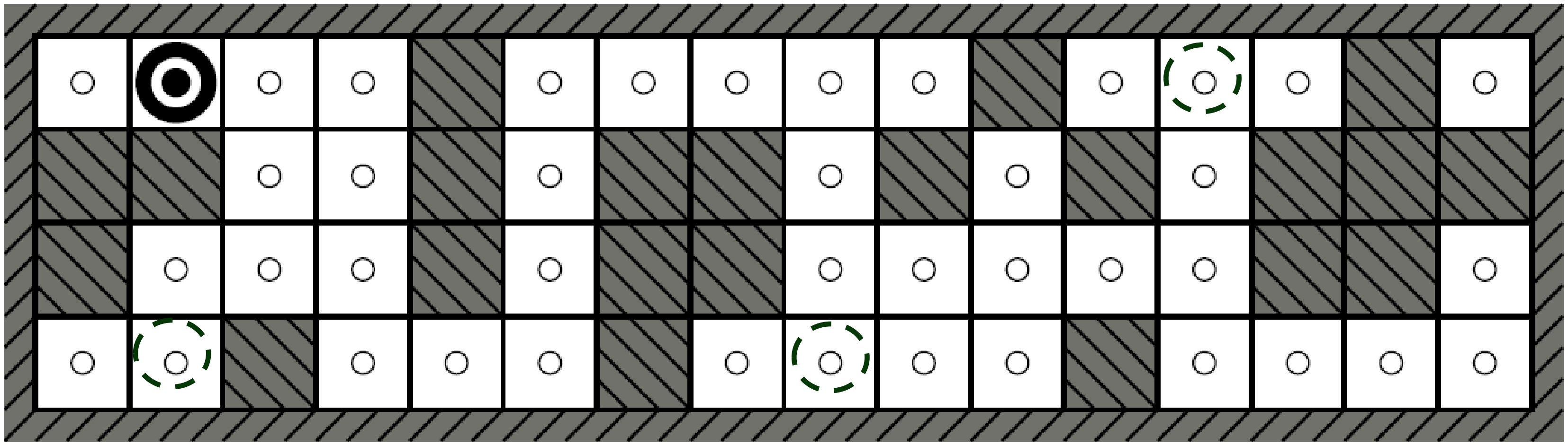


$NESW = 1010$

b_1 =Update$(b_a, 1010)$ // One possible location!



III. Online Search Agents

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Compute a complete solution before taking the first action.

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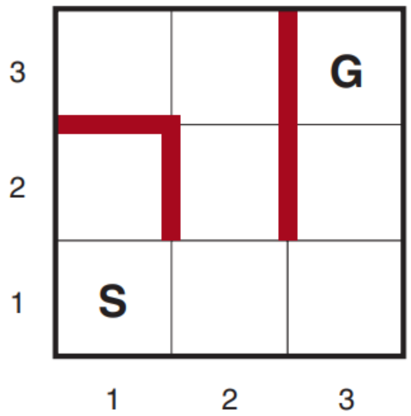

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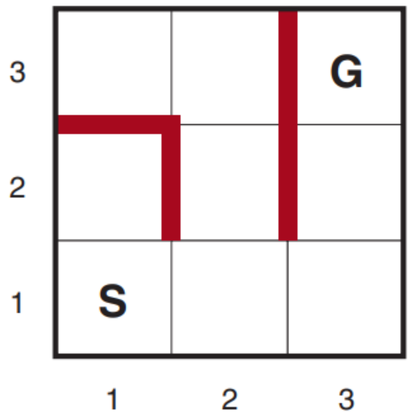

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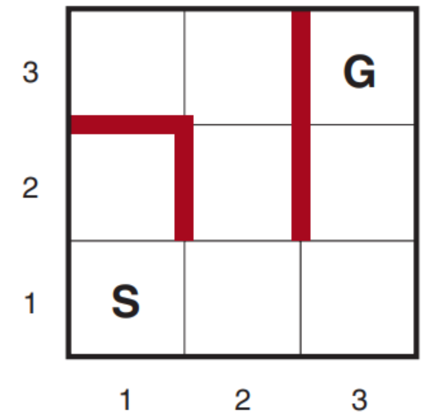
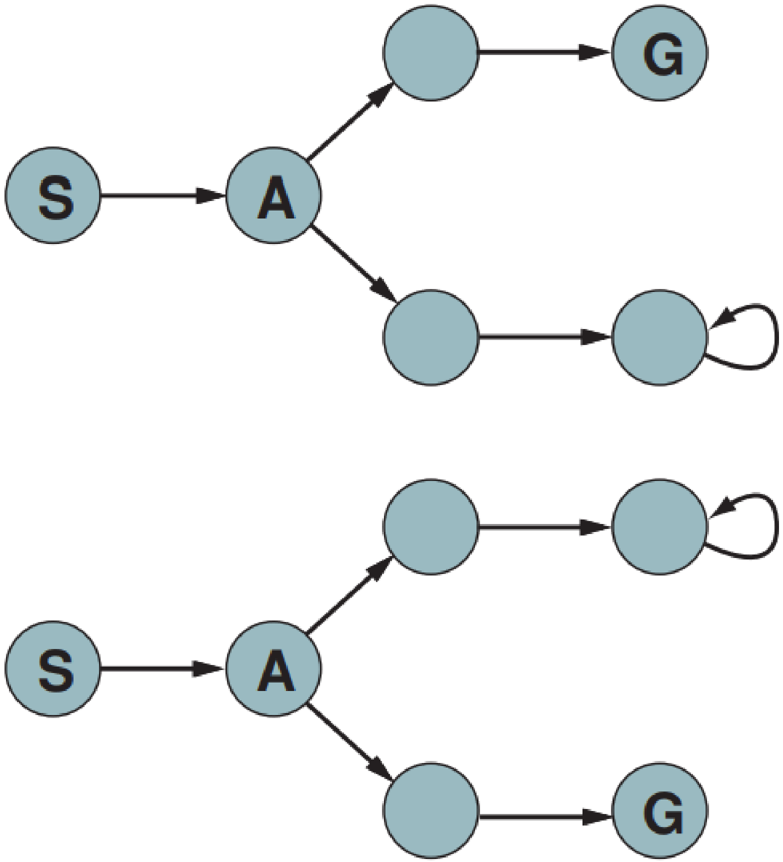


Unknown environment

- Good for search in dynamic or semi-dynamic environments.
- Allowing the agent to focus on contingencies that actually arise in nondeterministic domains.

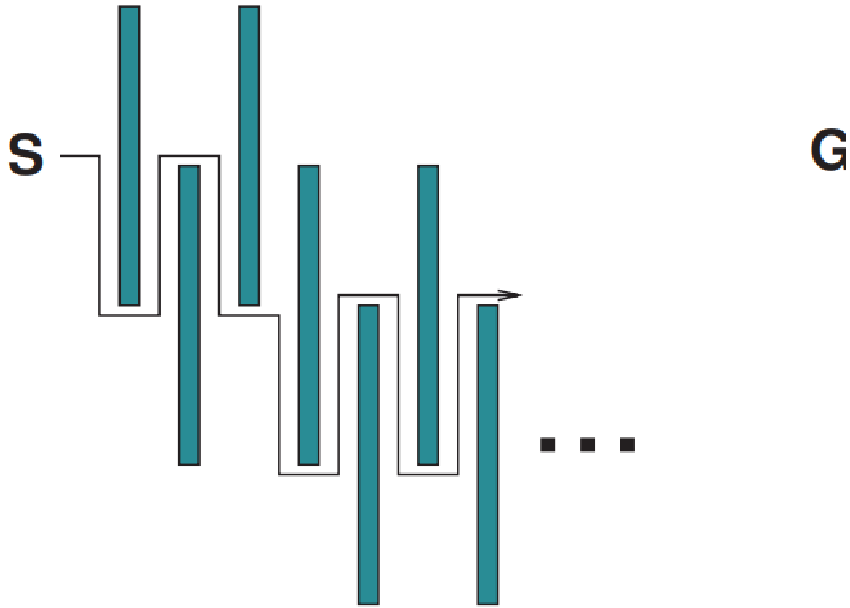
Dead Ends



Immediately after visiting S and A:

- ♠ Cannot tell whether it is in the top state space or the bottom one (because they look identical).
- ♠ Thus, no action exists to work in both state spaces.

Expensive Search Effort



As if the state space is being constructed by an imaginary adversary.

Whichever actions the agent chooses, dead ends and goals are placed “deliberately” to make the path inefficient.

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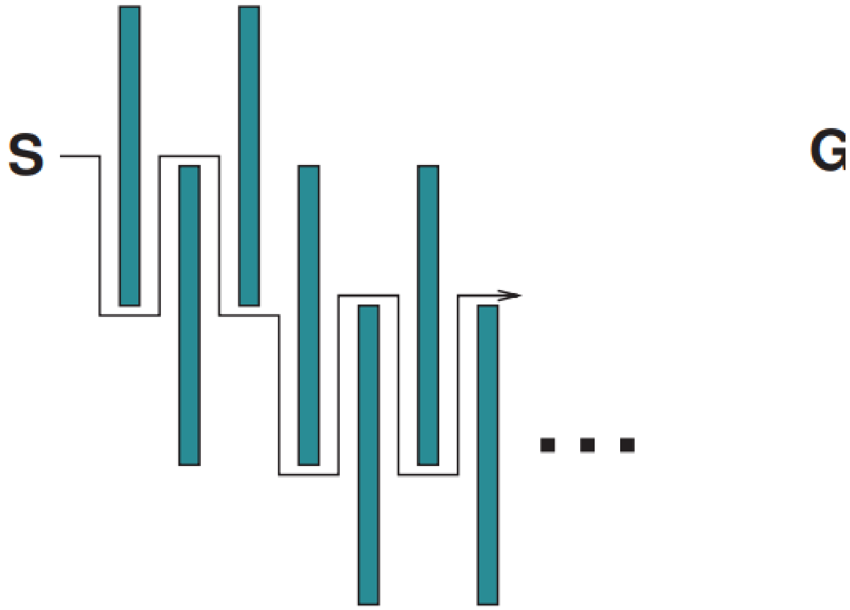

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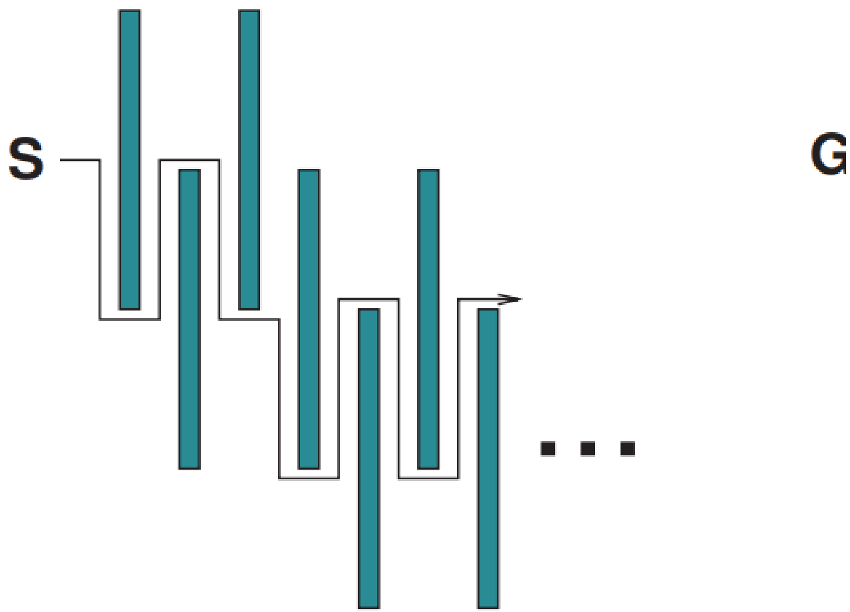

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Whichever actions the agent chooses, dead ends and goals are placed “deliberately” to make the path inefficient.

♠ The situation is complicated by that some actions are *irreversible*.

♦ Design a search agent that works in *safely explorable* state space.

Every reachable state can reach a goal state..